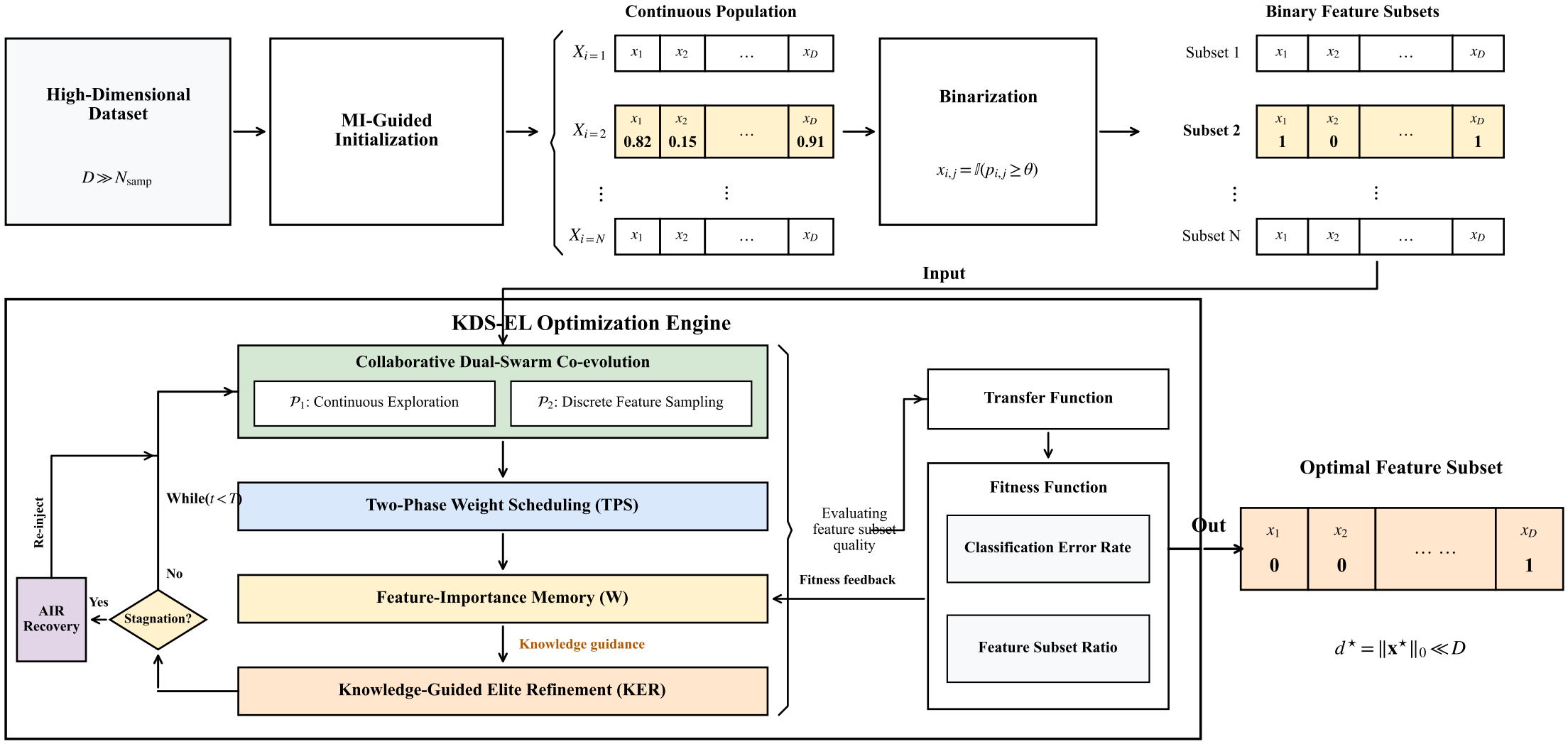

High-Dimensional Dataset
$D \gg N_{\text{samp}}$
MI-Guided Initialization
Continuous Population
$X_{i=1}$ x_1 x_2 … x_D
$X_{i=2}$ x_1 0.82 x_2 0.15 … x_D 0.91
$X_{i=N}$ x_1 x_2 … x_D
Binarization
$x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$
Binary Feature Subsets
Subset 1 x_1 x_2 … x_D
Subset 2 x_1 1 x_2 0 … x_D 1
Subset N x_1 x_2 … x_D
Input
KDS-EL Optimization Engine
Collaborative Dual-Swarm Co-evolution
$\mathcal{P}_1$: Continuous Exploration
$\mathcal{P}_2$: Discrete Feature Sampling
While($t < T$)
Two-Phase Weight Scheduling (TPS)
Feature-Importance Memory (W)
Knowledge guidance
Knowledge-Guided Elite Refinement (KER)
Re-inject
AIR Recovery
Yes
Stagnation?
No
Evaluating feature subset quality
Transfer Function
Fitness Function
Classification Error Rate
Feature Subset Ratio
Fitness feedback
Out
Optimal Feature Subset
x_1 0 x_2 0 … … x_D 1
$d^\star = \|\mathbf{x}^\star\|_0 \ll D$